

Molecular Biology
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Understanding life at the level of DNA, RNA, and proteins

Molecular Biology studies biological processes at the molecular level, focusing on DNA, RNA, and protein interactions that control how cells function and organisms develop.

● DEFINITION

The branch of biology concerned with the molecular basis of biological activity, particularly involving DNA, RNA, and protein synthesis.

● WHY IT MATTERS

Nearly all modern biotechnology, genetic medicine, and disease research rests on molecular biology's understanding of how genetic information is stored and expressed.

● WORKING PRINCIPLE



Molecular biologists study how genetic information encoded in DNA is transcribed into RNA and translated into proteins, and how this process is regulated to control cell behavior.

● QUICK FACTS

- ▶ The structure of DNA was discovered in 1953 by Watson and Crick, building on Rosalind Franklin's X-ray data
- ▶ The human genome contains about 3 billion DNA base pairs

● KEY TERMS



● CORE CONCEPTS

- ▶ Central dogma (DNA to RNA to protein)
- ▶ Gene expression regulation
- ▶ DNA replication
- ▶ Molecular cloning techniques

● APPLICATIONS

- ▶ Genetic disease diagnosis
- ▶ Gene therapy development
- ▶ Molecular diagnostic testing
- ▶ Genetic engineering

● ADVANTAGES

- ▶ Foundation for precision and genetic medicine
- ▶ Enables powerful diagnostic techniques
- ▶ Drives major advances in biotechnology

CAREER OPPORTUNITIES

Molecular biologist, Genetic researcher, Biotechnology scientist, Clinical geneticist

RESEARCH AREAS

Gene regulation, CRISPR gene editing, RNA biology

● REAL-LIFE EXAMPLES

- ▶ PCR testing for genetic and infectious diseases
- ▶ Gene therapy for inherited disorders
- ▶ DNA fingerprinting in forensics

● LIMITATIONS

- ▶ Gene regulation is highly complex and not fully understood
- ▶ Ethical questions around genetic manipulation
- ▶ Techniques can require expensive lab equipment

INDUSTRY APPLICATIONS

Pharmaceuticals, Diagnostics, Agriculture biotechnology, Forensic science

FUTURE SCOPE

Advances in gene editing and single-cell analysis are expected to deepen understanding of molecular disease mechanisms.

● CURRENT TRENDS

Rapid growth of single-cell molecular analysis techniques revealing cell-level genetic differences within tissues.

● SUMMARY

Molecular biology explains how DNA, RNA, and proteins control life at the cellular level, underpinning modern genetic medicine and biotechnology.